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Malicious Address Identifier (MAI): A Browser Extension to Identify Malicious URLs

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***Abstract*—Many websites have emerged since the internet was introduced. Every day, people visit a variety of websites, and they unknowingly click on malicious URLs because they simply neglect to check the URLs they visit. Clicking on a malicious URL has significant consequences. People might become victims of a phishing attack, which aims to steal people's money or identities by tricking them into disclosing their personal data such as passwords, credit card numbers, and bank account numbers on websites that portray themselves as legitimate. They might also have malware automatically installed on their computer devices, which can harm and destroy their systems. Therefore, if people don't know whether a URL is malicious or not, something more dangerous might happen. In this paper, we present a browser extension that employs a two-layer model to identify malicious URLs on the web using Bi-LSTM and CNN algorithms. This extension not only identifies malicious URLs but also prevents them from being visited. Furthermore, it blocks ads so that users do not click on malicious URLs through ads.**

***Index Terms*—Browser Extension, Bi-LSTM, CNN, Deep Learning, Malicious URL Detection, Web Attacks, Web Browser, Web Security**

I. Introduction

Web browsers are now widely used by a significant number of people to visit numerous websites on the Internet. In a single day, the Google Search Engine scans around 20 billion URLs, and according to a study, one-third of the URLs in use today are malicious [1]. The purpose behind generating these malicious URLs is often terrible, such as advancing a political agenda, stealing confidential information from businesses, or making quick money. Hackers frequently use popular search terms and viral videos to disseminate malicious software or links to phishing websites that operate maliciously on users' systems or trick them into revealing their personal data online [2].

Trojan horses, ransomware, worms, and spyware are examples of malicious software that targets people's computers. These viruses ultimately aim to damage the user's device, access personal information, and make money. Moreover, they could damage the network of the business, which would cost money. Although the objectives and methods used vary, the common purpose is to lure unknowing people to these malicious sites, which can be reached by links in emails, search engine results, or website redirection [3].

To identify malicious URLs, passive detection techniques often use lexical and host-based characteristics. Lexical characteristics are those that differ in appearance from actual URLs. For instance, the malicious URL www.paypal.com.pay.net has more "." than the legitimate URL www.paypal.com [4]. MAI uses a two-layer model to identify malicious URLs on the internet using Bi-LSTM and CNN algorithms that, in addition to identifying malicious URLs, prohibit their access and filter ads to prevent users from clicking on them via ads.

II. Problem Statement

Reason Behind the Project: With more websites being created daily, malicious websites are also becoming more prevalent in today's world. Unknowingly, people visit these

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websites, making it simple for hackers to target them.

These unknown clicks can occur for various reasons, including:

A. Reason-1 (due to the similar-looking URL)

The structure of the malicious URL differs only slightly from that of the real one. People are therefore unable to tell the difference between the authentic URL and its imitation. In fact, the user interfaces of the two websites are remarkably similar, and the imitation also attempts to entice visitors with appealing discounts or offers.

B. Reason-2 (due to advertisement)

Many advertisements appear while people are browsing the internet. Some advertisements may contain links to malicious websites or include malicious software. They pop up and draw attention by displaying content that is relevant to the viewer's interests and pretends to be legitimate.

C. Reason-3 (due to redirection)

When people browse the internet, they click on multiple URLs, but occasionally they might be redirected to another location or a website that they don't want to visit, or they might click on the URL accidentally. Along with clicking URLs, people also frequently click on images, videos, and other types of media that may be hyperlinked with malicious URLs and endanger themselves.

III. Related Work

The following approaches were identified in the literature after evaluating a number of research papers:

A. Comparative Study:

In order to compare the effectiveness of multiple well-known machine learning classifiers, the URLs were categorized as benign or dangerous as part of a binary classification problem. A public dataset of 450,000 URLs from Kaggle was used to train the model, out of which 23.2% (104,438) were malicious URLs and the rest were benign URLs. Machine learning classification algorithms such as Logistic Regression, Stochastic Gradient Descent, Random Forest, Support Vector Machine (SVM), Naive Bayes, K-Nearest Neighbor (KNN), and Decision Tree were used, and on the OpenPhish website, the most accurate classifier was used to find malicious URLs. Therefore, it turns out that the Random Forest achieves the maximum accuracy and FI score [5].

B. Multi-Layer Model:

Using a multi-layer model, the URLs were categorized as malicious or normal. The model primarily comprises four-layer classifiers, which are referred to as filters in this model. Therefore, the model comprises a four-layer filter composition. The filter can quickly determine whether a URL is malicious or not by training each layer's threshold as it approaches the threshold. If not, the filter hands the URL over to the next layer. Machine learning classifiers such as Naive Bayes, Decision Tree, and SVM were used to train the model, using a dataset of 20,000 samples, a dataset of 10,000 malicious URLs from the malicious website lab, and a dataset of 10,000 normal URLs from the first category directory [6].

C. Attention Based BiLSTM Model:

An AB-BiLSTM model based on deep learning was used to identify malicious URLs. The model automatically picks up new features, and while analyzing URL sequences, it employs an attention mechanism to capture anomalous segments. First, the URLs were preprocessed and turned into word vectors using pre-trained Word2Vec. Next, BiLSTM was trained in conjunction with an attention mechanism to extract features from URL sequences and classify them. The labeled URL dataset of 453,340 URLs, of which 344,821 are benign URLs from DMOZ and 108,519 are malicious URLs from VirusTotal and PhishTank, was used to test the model The experimental findings demonstrate that the suggested model is capable of achieving an accuracy rate of 98.06% and an F1 score of 95.22% [7].

D. K-Means Clustering:

Malicious URLs were grouped using K-Means clustering utilizing a dataset obtained from an open source threat intelligence feed framework comprising more than 11,000 malicious URLs. The dataset of malicious URLs contains hidden structures that were found using the unsupervised learning technique. Clustering of URL keywords reveals important information regarding malicious URLs. The clusters generated using the method have a silhouette coefficient of 0.383. Finally, in order to determine the percentage of malicious keywords included in a specified URL, a probabilistic scoring model was created. The model correctly recognised that more than 80% of the URLs in a test sample were malicious after analyzing more than 72,000 malicious keywords [8].

E. Convolutional Neural Network (CNN) Model:

Phishing URLs were identified based on the URL content. Online training and online testing make up the two components of the neural network-based approach to identifying malicious URLs. In order to access the page content, the simple features were first utilized for preliminary screening before being applied to the URLs that couldn't be distinguished by them. The approach based on CNN is then used to identify malicious URLs based on the content of the page. Another key element in identifying malicious websites is the webpage image. Therefore, a novel CNN-based method was also created to detect malicious URLs through the analysis of website images [9].

IV. Methodology

The current tab URL is collected by the MAI, which then sends the URL to the model to determine whether it is malicious or not. When MAI is on, it continuously checks the URL of the active tab and forbids people from accessing it if it is found malicious. MAI can be enabled and disabled as

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needed. Additionally, MAI stops advertisements from showing up on the webpage, shielding users from clicking on malicious URLs and falling for clickbait. Fig. 1 depicts the overall procedure.

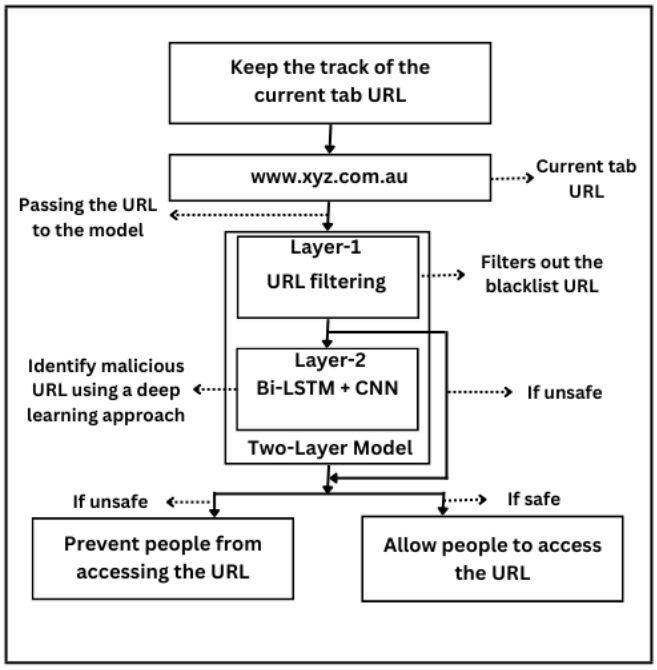

Fig. 1. MAI Overall Approach

The MAI model has two layers, with the first layer recognising the malicious URL using the conventional blacklisting method. If the URL is not present in the array of URLs that have been blacklisted, the model moves on to the next layer after searching the array. To identify the malicious URLs, the second layer employs Bi-LSTM and CNN algorithms. Finally, if the URL is malicious, it prevents anyone from viewing it.

The MAI model's two layers are described in detail below:

A. Layer-1: URL Filtering

The conventional blacklisting technique is used in the first layer of the MAI model to find malicious URLs online. It looks for the URL in the list of URLs that have been blacklisted, and if it finds the URL there, it prevents people from accessing it. Otherwise, it passes the URL on to the next layer for additional scrutiny, as shown in Fig. 2.

B. Layer-2: Bi-LSTM + CNN

The skip-gram pre-training model provides the initial embedding representation of the URLs for the second layer of the MAI model. Then, context dependency is extracted using Bi-LSTM to further understand the deep representation of URLs. Furthermore, CNNs are used to extract the local n-gram characteristics [10], as shown in Fig. 3.

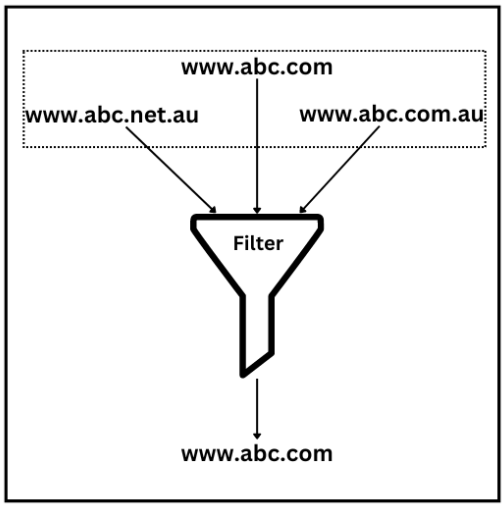

Fig. 2. URL Filtering

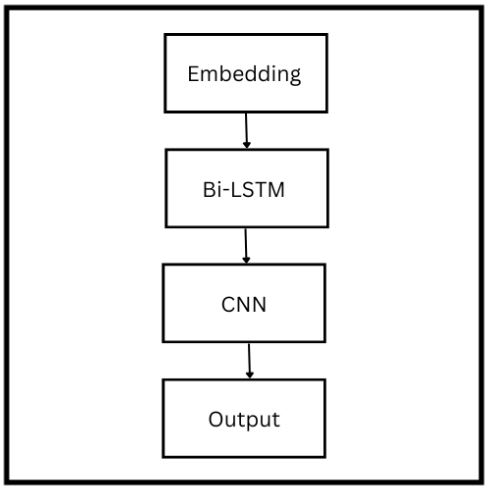

Fig. 3. MAI Layer-2 Architecture

IMPLEMENTATION

Fig. 4 illustrates the flow of the second layer of the MAI, and the subsequent explanation of each layer's functioning and the actions taken by it is described below:

a) InputLayer: Artificial input neurons make up this layer, which brings in the first batch of information that artificial neurons in later levels will process.

b) Embedding: Dense vectors are used to represent words, with each vector representing the word's projection into a continuous vector space.

c) Bidirectional(LSTM): It is used to better understand the deep representation of URLs, and context dependency is extracted. It significantly increases the quantity of information that is available to the network, which enhances the context that the algorithm can use.

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d) Conv1D: In order to produce a tensor of outputs, this layer creates a convolution kernel to capture local n-grams characteristics that are convolved with the layer input across a single spatial dimension.

e) BatchNormalization: The technique used to normalize the inputs to the layers by re-centering and re-scaling, speeding up and improving the stability of artificial neural network training.

f) Activation: The convolutional layer's activation function makes the output more nonlinear. The ReLu function is employed in this model as an activation function in a convolution layer.

g) MaxPooling1D: The maximum pooling operation is used for 1D temporal data. By picking the maximum value across a spatial window with a pool-sized resolution, the input representation is downsampled. Strides are used to move the window.

h) Flatten: Used to create a single, extended, continuous linear vector from all of the 2-Dimensional arrays generated by the pooled feature maps.

i) Dense: Simple layer of neurons, where each neuron receives input from every neuron in the layer above.

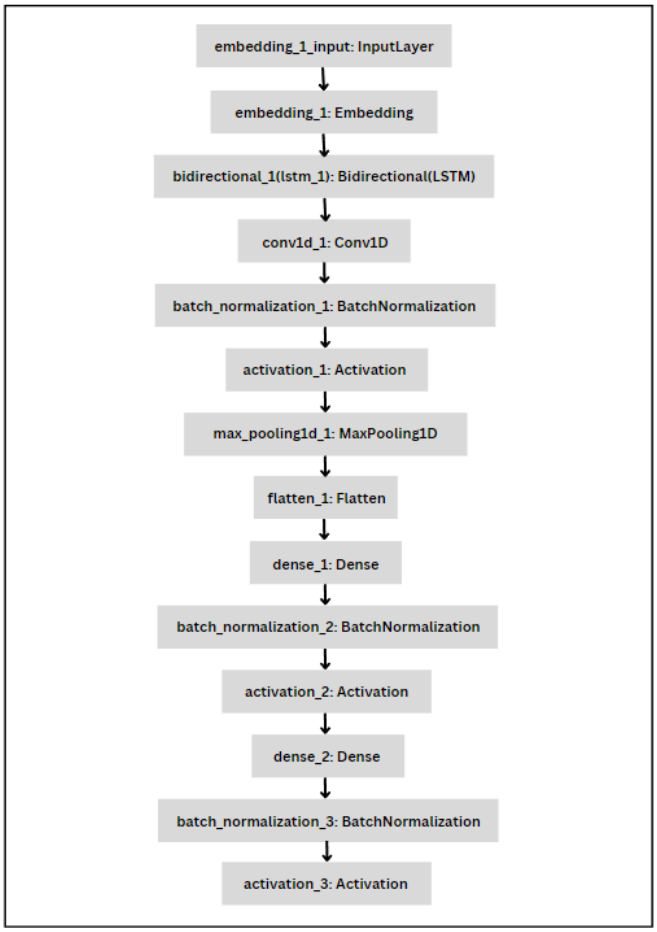

Fig. 4. MAI Layer-2 Flowchart

RESULT

MAI identifies the malicious URLs with 99.59% accuracy using Bi-LSTM and CNN algorithms. The model's training and validation accuracy are depicted in Fig. 5.

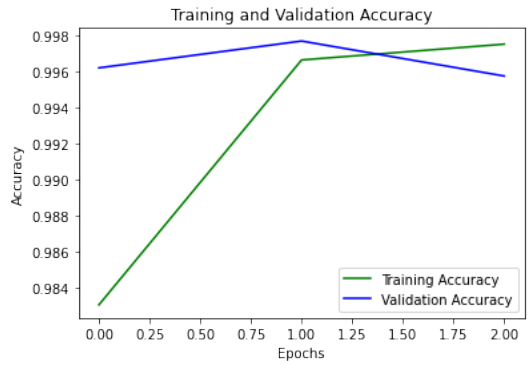

Fig. 5. Training and Validation Accuracy

SOCIAL IMPACT

A lot of people who neglect to double-check the URLs they visit benefit from MAI because it stops them from viewing malicious URLs on the internet. Now, users can browse the web without bothering about checking the URL.

The URL may often be hyperlinked to an image or other piece of media, making it invisible to people. Such hyperlinks can be found all over the Internet. Every day, people who use the internet for browsing come across numerous hyperlinked pieces of media. MAI also keeps an eye on redirection and prevents people from falling for the clickbait on the internet.

Clickbait attracts people by displaying content that they are interested in or by using fake news to advance an agenda [11]. Furthermore, websites like these are sometimes used for terrorist activities to transfer information in various languages [12] [13] [14]. MAI protects people from visiting such websites.

APPLICATIONS

The applications of MAI include:

1) World Wide Web: The World Wide Web (WWW) is an interconnected network fueled by the collection of links found throughout the Internet. The Internet link may be malicious and deceive unknowing people. MAI extension will protect such people from clicking malicious links.

2) Social Media: Social media is widely used today, and many rumors are disseminated to mislead people. People are drawn to these rumors and may click on them, which may lead to malicious websites. Where they could disclose their personal information [15]. The MAI model can be integrated with social media to check for such links.

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3) Email: Malicious links that deceive or harm people can also be found in emails. Email can be connected to the MAI model to scan every message for malicious links and prevent recipients from opening them.

CONCLUSIONS

The number of malicious websites and online crime are both rising. MAI safeguards people against being duped by malicious URLs. Stopping people from accessing the malicious URLs and blocking the advertisements. It uses a two-layer model and is simple to use and manage in order to identify malicious URLs.

REFERENCES

[1] A. Das, A. Das, A. Datta, S. Si and S. Barman, "Deep Approaches on Malicious URL Classification," 2020 11th International Conference on Computing, Communication and Networking Technologies (ICCCNT), 2020, pp. 1-6, doi: 10.1109/ICCCNT49239.2020.9225338.

[2] Y. -C. Chen, Y. -W. Ma and J. -L. Chen, "Intelligent Malicious URL Detection with Feature Analysis," 2020 IEEE Symposium on Computers and Communications (ISCC), 2020, pp. 1-5, doi: 10.1109/ISCC50000.2020.9219637.

[3] G. Chakraborty and T. T. Lin, "A URL address aware classification of malicious websites for online security during web-surfing," 2017 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS), 2017, pp. 1-6, doi: 10.1109/ANTS.2017.8384155.

[4] R. K. Nepali and Y. Wang, "You Look Suspicious!!: Leveraging Visible Attributes to Classify Malicious Short URLs on Twitter," 2016 49th Hawaii International Conference on System Sciences (HICSS), 2016, pp. 2648-2655, doi: 10.1109/HICSS.2016.332.

[5] Shantanu, B. Janet and R. Joshua Arul Kumar, "Malicious URL Detection: A Comparative Study," 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS), 2021, pp. 1147-1151, doi: 10.1109/ICAIS50930.2021.9396014.

[6] R. Kumar, X. Zhang, H. A. Tariq and R. U. Khan, "Malicious URL detection using multi-layer filtering model," 2017 14th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP), 2017, pp. 97-100, doi: 10.1109/ICCWAMTIP.2017.8301457.

[7] F. Ren, Z. Jiang and J. Liu, "A Bi-Directional LSTM Model with Attention for Malicious URL Detection," 2019 IEEE 4th Advanced Information Technology, Electronic and Automation Control Conference (IAEAC), 2019, pp. 300-305, doi: 10.1109/IAEAC47372.2019.8997947.

[8] S. Nayak, D. Nadig and B. Ramamurthy, "Analyzing Malicious URLs using a Threat Intelligence System," 2019 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS), 2019, pp. 1-4, doi: 10.1109/ANTS47819.2019.9118051.

[9] Y. Chen, Y. Zhou, Q. Dong and Q. Li, "A Malicious URL Detection Method Based on CNN," 2020 IEEE Conference on Telecommunications, Optics and Computer Science (TOCS), 2020, pp. 23-28, doi: 10.1109/TOCS50858.2020.9339761.

[10] L. Zhang and P. Zhang, "PhishTrim: Fast and adaptive phishing detection based on deep representation learning," 2020 IEEE International Conference on Web Services (ICWS), 2020, pp. 176-180, doi: 10.1109/ICWS49710.2020.00030.

[11] A. Yerlekar, N. Mungale and S. Wazalwar, "A multinomial technique for detecting fake news using the Naive Bayes Classifier," 2021 International Conference on Computational Intelligence and Computing Applications (ICCICA), 2021, pp. 1-5, doi: 10.1109/ICCICA52458.2021.9697244.

[12] R. S. Dudhabaware and M. S. Madankar, "Review on natural language processing tasks for text documents," 2014 IEEE International Conference on Computational Intelligence and Computing Research, 2014, pp. 1-5, doi: 10.1109/ICCIC.2014.7238427.

[13] Nagdive AS, Tugnayat R, Peshkar A. Int. J. Eng. Adv. Technol. 2020; 9(3): 2553-2559.

[14] A. Kaur and U. Shrawankar, "Adverse conditions and techniques for cross-lingual text recognition," 2017 International Conference on Innovative Mechanisms for Industry Applications (ICIMIA), 2017, pp. 70-74, doi: 10.1109/ICIMIA.2017.7975565.

[15] P. Maske, K. Pardhi, S. Hatzade, R. Sharma and S. Wazalwar, "Image Security Barrier (ISB): Hide valuable information in image using machine learning," 2022 10th International Conference on Emerging Trends in Engineering and Technology - Signal and Information Processing (ICETET-SIP-22), 2022, pp. 1-6, doi: 10.1109/ICETET-SIP-2254415.2022.9791749.

[16] S. K. Tiwari, P. M. Girade, B. P. Khobe, A. Rambhajani, S. Wazalwar and M. Khanapurkar, "The Novel Approach of QR Code detection with Smart Glasses," 2022 10th International Conference on Emerging Trends in Engineering and Technology - Signal and Information Processing (ICETET-SIP-22), 2022, pp. 01-05, doi: 10.1109/ICETET-SIP-2254415.2022.9791569.

[17] Wazalwar, S.S., Shrawankar, U. (2023). Sign Language versus Spoken English Language—A Study with Supervised Learning System Using RNN for Sign Language Interpretation. In: Tiwari, R., Pavone, M.F., Ravindranathan Nair, R. (eds) Proceedings of International Conference on Computational Intelligence. Algorithms for Intelligent Systems. Springer, Singapore.

[18] Sampada S. Wazalwar, Urmila Shrawankar,(2017). Interpretation of sign language into English using NLP techniques, Journal of Information and Optimization Sciences, 38:6, 895-910, DOI: 10.1080/02522667.2017.1372136.

[19] S. S. Wazalwar and U. Shrawankar, "Multimodal Interface for Deaf and Dumb Communication," 2019 6th International Conference on Computing for Sustainable Global Development (INDIACom), 2019, pp. 418-422.

[20] P. Patankar, S. Dorle, N. Wyawahare and L. P. Thakre, "Comparative Study on Design Of AI-Based Communication Protocol For VANET," 2022 IEEE 4th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA), Goa, India, 2022, pp. 451-455, doi: 10.1109/ICCCMLA56841.2022.9989247.

[21] P. Malewar, R. Kadu, R. Kakde, N. P. Wyawahare and R. Agrawal, "A Review Paper on Data Sensing and Acquisition Complexity and Accuracy in Green House Monitoring Systems," 2022 10th International Conference on Emerging Trends in Engineering and Technology - Signal and Information Processing (ICETET-SIP-22), Nagpur, India, 2022, pp. 1-6, doi: 10.1109/ICETET-SIP-2254415.2022.9791732.

[22] A. K. Dudhpachare, T. V. Kuthe, C. V. Lalke, N. P. Wyawahare, R. Agrawal and P. Daigavane, "Process of RO's Waste Water Reuse Water Management in Society by Using IOT Automation," 2022 International Conference on Applied Artificial Intelligence and Computing (ICAAIC), Salem, India, 2022, pp. 1546-1550, doi: 10.1109/ICAAIC53929.2022.9792805.

[23] S Wazalwar, U Shrawankar, Online Healthcare Consultation System for Deaf and Dumb During Pandemic Situation, Bioscience Biotechnology Research Communications, Special Issue Vol 13 No 14 (2020) Pp-213-216,Doi: http://dx.doi.org/10.21786/bbrc/13.14/50.

[24] Wazalwar, S.S., Shrawankar, U. (2021). Distributed Education System for Deaf and Dumb Children and Educator: A Today's Need. In: Singh Mer, K.K., Semwal, V.B., Bijalwan, V., Crespo, R.G. (eds) Proceedings of Integrated Intelligence Enable Networks and Computing. Algorithms for Intelligent Systems. Springer, Singapore.

[25] S Wazalwar, U Shrawankar , Community Cloud Service Model for People with Special Needs, Cloud Computing Technologies for Smart Agriculture and Healthcare, 47-55, Chapman and Hall/CRC, 2021

[26] N. A. Karotiya, N. P. Wyawahare and S. L. Haridas, "Review paper on point to point communication with the use of power over Ethernet based on VOIP system on asterisk," 2016 2nd International Conference on Advances in Electrical, Electronics, Information, Communication and Bio- Informatics (AEEICB), Chennai, India, 2016, pp. 729-732, doi: 10.1109/AEEICB.2016.7538390.

[27] A. P. Tembhurne and N. P. Wyawahare, "Alamouti based on spatial modulation for data transmission," International Conference for Convergence for Technology-2014, Pune, India, 2014, pp. 1-4, doi: 10.1109/I2CT.2014.7092179.

[28] S. Patil, S. Vairagade and D. Theng, "Machine Learning Techniques for the Classification of Fake News,"2021 International Conference on Computational Intelligence and Computing Applications (ICCICA), Nagpur, India, 2021, pp. 1-5, doi: 10.1109/ICCICA52458.2021.9697267.

[29] A. Khan and A. V. Sakhare, "A Gestalt of Data Exploration and Conception," 2015 Fifth International Conference on Communication Systems and Network Technologies, Gwalior, India, 2015, pp. 908-912, doi: 10.1109/CSNT.2015.240.

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